Pranav Gangwar

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Summary

PhD Candidate seeking a full-time Spring 2027 role in computer architecture and performance modeling, with experience in GPU acceleration, kernel optimization, and parallel algorithm design across CPU, GPU, and PIM.

Education

2021 – Present	PhD in Computer Engineering at **UC San Diego**	(GPA: 4.0)
2021 – 2024	MS in Computer Engineering at **UC San Diego**	(GPA: 4.0)

Relevant Coursework: Principles of Computer Architecture, Principles of Operating Systems, Low-power VLSI for Machine Learning, Parallel Computation

Work Experience

GPU Performance Architecture Intern — **Intel** (Jun-Sept 2023, Jun-Sept 2022)

- Evaluated next-generation Tensor Core architecture proposals across diverse ML workloads using an analytical performance simulator, identifying memory bottlenecks and recommending architectural improvements.
- Developed a matrix tile-size selection algorithm for Intel GPUs, improving Tensor Core utilization by reducing memory stalls and redundant computation across ML workloads.
- Optimized a state-of-the-art matrix decomposition algorithm for Intel GPUs, improving single-GPU utilization on ML workloads.
- Developed a performance model for NeRF to evaluate the proposed GPU architecture.

Digital Design Engineer — **Texas Instruments** (July 2018 - July 2021)

- Led complete physical design from floorplanning through sign-off, providing architectural feedback to improve pipelining and meet stringent I/O timing constraints.
- Developed scripts for automated, area-efficient FIR filter RTL generation.
- Developed a leakage power recovery flow by resizing standard cells at sign-off, leveraging reduced timing pessimism relative to place-and-route.

Course Projects

Fused Multi-Head Latent Attention (MLA) CUDA Kernel

- Implemented a CUDA kernel for Multi-Head Latent Attention (DeepSeek-V3-style low-rank KV-compression attention), achieving ∼**0.83x of PyTorch (cuBLAS) throughput** across **Prefill and Decode scenarios**.

GPU Accelerated General Matrix Multiplication (GEMM)

- Implemented high-performance CUDA GEMM kernels (shared-memory tiling, coalesced global memory access, register-blocked ILP), achieving ∼**0.68x of cuBLAS throughput** across **matrix sizes N=256–2049**.

Research Projects

Computer Architecture Research **- Advisor: Tajana Rosing**

TermiNETor: An Energy-Efficient DNN Accelerator — Sep 2021 - Jun 2022

- Achieved **120× energy efficiency over GPUs** by co-designing a novel bit-serial hardware architecture with an early-termination algorithm to prune redundant computations.
- Proposed the architecture and validated its performance gains using an analytical simulator, with RTL-based power and area estimation; presented at ***IEEE ICCD 2022***.

FHEmem: Near-Memory Data Processing Accelerator for FHE — Sep 2021 - Dec 2022

- Co-designed a DRAM-based near-data processing (NDP) architecture that performs memory-bound FHE primitives in memory, avoiding costly off-chip data movement.

- Contributed to the near-mat unit design for in-memory number-theoretic transform (NTT) butterfly operations, enabling FHEmem to achieve **4.0x speedup and 6.9x energy-delay-area efficiency over existing accelerators**; published in ***IEEE TETC (2025 Best Paper Runner-Up)***.

Bioinformatics Research - Advisor: Yatish Turakhia

High-Resolution Variant Detection from Wastewater & Clinical Samples Jan 2023 - Jan 2026

- Developed a **scalable**, **multi-threaded**, and **memory-efficient** algorithm (WEPP) that processes millions of genome fragments against a 16-million SARS-CoV-2 genome database in hours, accurately identifying complete genomes—solving a problem previously considered intractable.
- Enabled early detection of emerging variants for timely public health response; adopted by the CDC for nationwide surveillance and published in ***PLOS Computational Biology***.
- Generalized the approach (metaWEPP) to identify complete genomes of multiple co-occurring pathogens from wastewater or clinical DNA samples; published in ***NAR Genomics and Bioinformatics***.

SARS-CoV-2 Vaccine Selector Sep 2025 - Present

- Analyzed 16 million SARS-CoV-2 genomes and found that variant prevalence and existing *in silico* fitness estimators provide complementary signals for predicting future variant dominance.
- Designed a **lightweight ML algorithm** integrating variant prevalence and fitness estimates to select vaccine variants that provide broad protection against future infections, slightly outperforming FDA recommendations.

Publications

1. **TermiNETor: Early convolution termination for efficient deep neural networks**, U Mallappa, **P Gangwar**, B Khaleghi, H Yang, T Rosing. *IEEE Int. Conference on Computer Design (ICCD)*, 2022.
2. **FHEmem: A processing in-memory accelerator for fully homomorphic encryption**, M Zhou, Y Nam, **P Gangwar**, W Xu, A Dutta, et al. *IEEE Transactions on Emerging Topics in Computing*, 2025.
3. **WEPP: Phylogenetic Placement Achieves Near-Haplotype Resolution in Wastewater-Based Epidemiology**, **P Gangwar**, P Katte, M Bhat, Y Turakhia. *PLOS Computational Biology*, 2026.
4. **metaWEPP: Leveraging Biobank-Scale Intra-Species Phylogenies for Near-Haplotype Resolution in Metagenomic Analysis**, **P Gangwar**, Q Xu, J Seangmany, P Katte, Y Turakhia. *NAR Genomics and Bioinformatics*, 2026.
5. **Genetic Diversity Drives the Rate and Fitness Jumps of Detectable SARS-CoV-2 Recombination**, K Smith, **P Gangwar**, J Wertheim, Y Turakhia. Under Review *Nature Microbiology*.

Awards & Talks

- **Best Paper Runner-Up** — FHEmem, IEEE Transactions on Emerging Topics in Computing, 2025
- **Innovation in Research Award** — Microbes in Wastewater Symposium, 2026
- **Talks** — WEPP & metaWEPP, ISMB 2026

Skills

Languages	C++, C, CUDA, Verilog, Python, Tcl, Shell
Software & Tools	PyTorch, Innovus™, Tempus™, Voltus™, PVS, Assura®, Vivado®, MATLAB®